STRATIFICATION BY FIBRE DEGREE FOR FLAT, FINITELY PRESENTED, SEPARATED AND LOCALLY QUASI-FINITE MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 12904-12911, immediately following the statement at 12891-12902.

Editorial note. The proof reduces to S affine and then to quasi-compact opens W; both moves are displayed in the span, and only the union step is compressed into one sentence.

Editorial note. The cited chapter lemma is a navigation pointer into the same chapter.

Allowed standard prerequisites (editorial).

- Stratification by fibre degree in the quasi-finite case. chapter lemma lemma-stratify-flat-fp-qf, cited at proof line 12909. Supplies openness of the points where the fibres of a quasi-compact piece W -> S have degree at least d.
- Meaning of the degree of a fibre over a field-valued point. used at proof lines 12900-12901 and 12905-12906. Identifies factoring through U_d with the fibre X x_S Spec(k) having degree at least d over k.

Source reference. Original lemma 07RZ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12845–12845. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a flat, locally of finite presentation, separated, and locally quasi-finite morphism of schemes. Then there exist open subschemes*

$$S = U_0 \supset U_1 \supset U_2 \supset \ldots$$

such that a morphism $\mathrm{Spec}(k) \to S$ where k is a field factors through U_d if and only if $X \times_S \mathrm{Spec}(k)$ has degree $\geq d$ over k.

Proof. The statement simply means that the collection of points where the degree of the fibre is $\geq d$ is open. Thus we can work locally on S and assume S is affine. In this case, for every $W \subset X$ quasi-compact open, the set of points $U_d(W)$ where the fibres of $W \to S$ have degree $\geq d$ is open by Lemma 07RZ. Since $U_d = \bigcup_W U_d(W)$ the result follows. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.